

Senior Project

Team Charter

Holographic Alphabet Toy

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September 8, 2020

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Team Information

1. Team Name

Alien Alphabets

2. Motto

“Let’s start with A!”

3. Mascot

Alien

4. Vision

To create an everlasting interactive educational toy. The vision is to teach children that education is not an end goal. In order to ensure a better future for our children, we must encourage continuous learning and highlight the importance of critical thinking.

Mission/Purpose

Our mission is to enhance the critical thinking skills of our future leaders by using a fun and unique visual learning style, coupled together with critical thinking questions and real world information based on the appropriate age groups.

Goals

- Understanding hardware and knowledge requirements
- Gather all related content
- Create a model design
- Test basic hardware functions with Arduino board
 - Functional button
 - Display image
 - Functional menu
- Debugging
- Build the hardware design for each button
- Build a shell
- Integrate hardware and software
- Build a stand
- Have fun

Roles

1. Coordinator: sets agenda in advance, manages the work tasks of the meeting
 - Andy Nguyen
2. Scribe: records results of meetings
 - Sean Lovstuen
3. Timekeeper: tracks the time set for agenda items, help team keep on schedule
 - John Paul Baumann
4. Facilitator: helps people work together smoothly, helps people communicate and cooperate, checks to see that all points of view are expressed and everyone has a chance to participate
 - Dr. Perez-Davila
5. Team Members: contributes fair share, supports others, and works for the team

Hardware Team:

- John Paul Baumann
- Andy Nguyen

Software Team:

- Syed Nayeem
- Sean Lovstuen
- Khoi Pham

Ground Rules

7.8 IEEE Code of Ethics

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons and to not engage in acts of discrimination based on race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

Administrative:

- Where will we meet? How often? What time will we start?
 - Weekly discord meeting
 - Soft meeting schedule
- How will we monitor our progress?
 - Past: recap of the past week
 - Present: notable achievements of the week
 - Future: planning for future achievements

Task:

- How will we make decisions?
 - High majority vote
- How will we handle disagreement?
 - Understanding and communication/rock paper scissors
- What should be our guidelines regarding participation?
 - Everyone should contribute equally
- How will we do problem solving?
 - Break into parts and solve little by little

Interpersonal:

- Follow the IEEE standard on Code of Ethics

Procedures**What do I need to know beforehand?**

- Basic understanding of C/C++ syntax
- Complete understanding of what Arduino is and what it is used for

What can I do to make this project successful?

- Understand the objective
- Become familiar with Arduino IDE
 - Review functions for Arduino Board
 - Read through how to create a menu screen
 - Read through how to display images onto a screen using Arduino by pushing a button
- Become familiar with different types of Arduino hardware and how to use them
 - Breadboard(regular and mini)
 - Circuit wires
 - Resistors, etc.

- Create mental images of how everything should correspond with each other

1. **Objective:** Build an interactive children's toy that can continuously be updated with new content
2. **Requirements:**

The following is subject to change as needed

- Software:
 - Arduino IDE
- Hardware:
 - Arduino Mega 2560 Rev 3 (~\$40.00)
 - 8-bit board
 - 54 digital pins
 - 16 analog inputs
 - 4 serial ports
 - 5" TFT LCD Display (~\$40.00 - ~\$90.00)
 - Screen that will display hologram images
 - Arduino compatible
 - 3" TFT LCD Touchscreen Display (~\$20.00)
 - Buttons (~\$6.00 - \$13.00 for a pack)
 - 26 Buttons (One for each letter)
 - Mini Breadboards (~\$10.00 for 12 pack)
 - One for each button
 - 22 AWG stranded wires
 - To connect buttons to main breadboard
 - Speaker
- Toy requirements:
 - A menu screen
 - For choosing appropriate age group for learning
 - A holographic display
 - For images
 - 26 Alphabet buttons
 - For displaying letters and information
- Constraints:
 - Both hardware and software must be Arduino compatible
 - Toy must be able to be continuously updated
 - Questions and information must be given based on age group

3. Design:

- Create menu screen on 3” display
- Create 3d images to project off 5” display
- Create a hologram projector using clear hard plastic
- Connection between Arduino board, menu display, main display, and buttons

Team Self-Assessment Data

- Keep track of attendance
- Code commits
- Inputs into project ideas

Commitment

- **Andy Nguyen**
- **John Paul Baumann**
- **Khoi Pham**
- **Syed Nayeem**
- **Sean Lovstuen**
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